

Kyle Main

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Senior Machine Learning Engineer — Applied ML & GenAI Security

12 years building production ML systems and security detection content — clustering, time-series anomaly detection, and precision/false-positive-tuned models deployed against massive-scale data — plus hands-on experience putting GenAI/LLM tooling directly into live security workflows: prompt engineering for security triage, LLM-driven orchestration of SIEM APIs, and reusable GenAI-powered tooling built for other engineers. Full ML lifecycle ownership: problem definition, data pipelines/EDA at scale (PySpark/GCP Dataproc), model development (PyTorch/scikit-learn), production deployment, and continuous precision/recall tuning fed by real production outcomes.

CORE SKILLS

GenAI/LLM for Security: Prompt engineering for security triage and detection-content generation; GenAI-driven orchestration of SIEM APIs across 9 platforms; reusable GenAI-powered tooling built for other engineers (e.g., automated cross-SIEM detection-rule conversion) — real production LLM-into-workflow experience

ML & Applied Data Science: Python, SQL, pandas, scikit-learn, NumPy, SciPy, PyTorch, clustering/unsupervised ML, time-series anomaly detection, R

Data Pipelines & EDA at Scale: PySpark / SparkSQL, GCP Dataproc, Zeppelin notebooks, BigQuery, Apache Beam / GCP Dataflow, large-scale dataset construction across 220+ ingested sources

Model Evaluation & Production Rigor: Precision/recall and false-positive/false-negative analysis at production scale; formally tracked detection-quality metrics; staged/safe rollout before full production deployment; rollback paths

Engineering & Platform: Git, Docker, GitLab CI/CD (detection-as-code with automated testing), Kubernetes-orchestrated platform experience, Elasticsearch (queries, transforms, native detection rules, API)

PROFESSIONAL EXPERIENCE

Senior Cybersecurity Engineer — Shorepoint

Oct 2023 – Present

- Built production GenAI tooling for security automation: prompt engineering for false-positive triage, automated detection-content generation, and reusable GenAI-powered "skills" that convert detection rules across SIEM syntaxes for other engineers.
- Currently own detection/alerting analytics (Splunk saved searches) for Treasury's SOC, translating false-positive/false-negative investigation findings directly into refined detection logic (TSSOC, current project).
- Built an entirely new Elasticsearch-based security data platform from scratch for DOE/NNSA — including a UEBA detection layer on custom data transforms modeling behavioral baselines from CrowdStrike, Suricata, and Zeek telemetry (completed).

Senior Threat Detection Engineer and Data Scientist — Forescout

Aug 2022 – Oct 2023

- Senior member of a data science and detection engineering team building signature, statistical, behavioral, time-series, and ML-based detection content against massive-scale customer data on a cloud-based big-data platform.

Threat Detection Engineer and Data Scientist — Trend Micro / Cysiv

Sep 2018 – Aug 2022

- Built ML-based detection content clustering devices on the network by behavioral features (unsupervised learning), and designed time-series anomaly detection models for entity behaviors — process chains, parent/child process patterns, authentication volume/geography — as part of 2,300+ detection rules covering the MITRE ATT&CK matrix.
- Ran exploratory data analysis at scale on GCP Dataproc via PySpark/SparkSQL to validate new detection signal candidates before productionizing; owned data engineering for 220+ ingested log sources feeding the detection pipeline.

Security Data Scientist — Experian

Jan 2015 – Jan 2018

- Analyzed large-scale security datasets to build custom models and algorithms identifying emerging threats — DNS-based malware detection/mitigation and anomalous-behavior discovery in log data.

EDUCATION & CERTIFICATIONS

M.S. Physics — University of North Texas (2013) | B.S. Physics — Ball State University (2011)

Splunk User Certification · Splunk for Analytics and Data Science · Johns Hopkins (Coursera): R Programming, The Data Scientist's Toolbox